

Geographic Information Systems

Week-10: Assignment-10

- 1 Which of the following cannot be modelled using a DEM?
 - (a) Slope
 - (b) Aspect
 - (c) Geology**
 - (d) Runoff
- 2 The rate of change of elevation is called:
 - (a) Gradient
 - (b) Slope**
 - (c) Aspect
 - (d) Gradient slope
- 3 What is the difference between slope and aspect?
 - (a) Slope is the distance down the fall line from the top of the slope to its bottom, while aspect is the percentage gradient of this line averaged over its full distance.
 - (b) Slope is the gradient directly down the fall line, while aspect is the direction of the fall line relative to north.**
 - (c) Slope is the direction of the fall line, while aspect is the gradient of the fall line.
 - (d) Slope is the gradient of the fall line relative to vertical, while aspect is the direction of the fall line relative to the line of greatest slope.
- 4 Slope can be calculated from the formula $S = b^2 - c^2$.
 - (a) True**
 - (b) False
- 5 What is meant by the term 'precision'?
 - (a) The extent to which a value approaches its true value
 - (b) The lack of bias in the data
 - (c) The level of detail at which data is stored**
 - (d) The overall quality of the data
- 6 What is meant by the term 'data quality'?
 - (a) The lineage of the data
 - (b) The generalization present in the source data
 - (c) The resolution of the data
 - (d) The inherent quality of the data as characterized by its accuracy, precision, bias, level of error, etc.**
- 7 When an error in a dataset leads to the commission of another error this is called error:
 - (a) Propagation**
 - (b) False precision
 - (c) Horizontal error
 - (d) Cascading
- 8 DEMs can be prepared from:
 - (a) Raster stereo pair

- (b) Contours
- (c) InSAR technique
- (d) All the above**

9 The shape of unit of DEM can only be:

- (a) Rectangular
- (b) Circular
- (c) Square**
- (d) Triangle

10 The cell value of a DEM can have:

- (a) Both positive and negative, integer and real values**
- (b) Only positive integer values
- (c) Only negative integer and real values
- (d) Only positive real values

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